

ICT Fundamentals

Student Learning Outcomes:

After the completion of this unit students will be able to:

- ◆ Identify the various emerging technologies and their usage, such as Artificial Intelligence, Robotics, Biometrics, Computer-Assisted Translation, Virtual Reality, 3D and holographic imaging, Cloud Computing.
- ◆ Understand the progression in Computing Technologies.
- ◆ Recognize different peripherals of the computer system and their applications.
- ◆ Identify different I/O ports and functions of I/O ports in computer system.

Introduction Of Unit:

This unit covers the basic concept of emerging technologies such as (Artificial Intelligence, Robotics, biometrics and etc). This unit also gives a brief overview of the progression in computing technologies, different types of I/O ports and various peripheral devices.

1.1 Emerging Technologies:

Emerging technologies are new ideas still growing. They are like baby technologies. These new tools can:

- ◆ Solve big problems (like cleaning dirty water).
- ◆ Make new jobs (like robot helpers).
- ◆ Make life better (like smart watches that tell time and health).

Examples: Artificial Intelligence (AI), robots, Virtual Reality (VR) glasses, and cloud storage. A caller uses voice assistants (e.g., Siri, Google Assistant) on phones is an example of every day usage. Figure 1.1 shows some of emerging technologies.

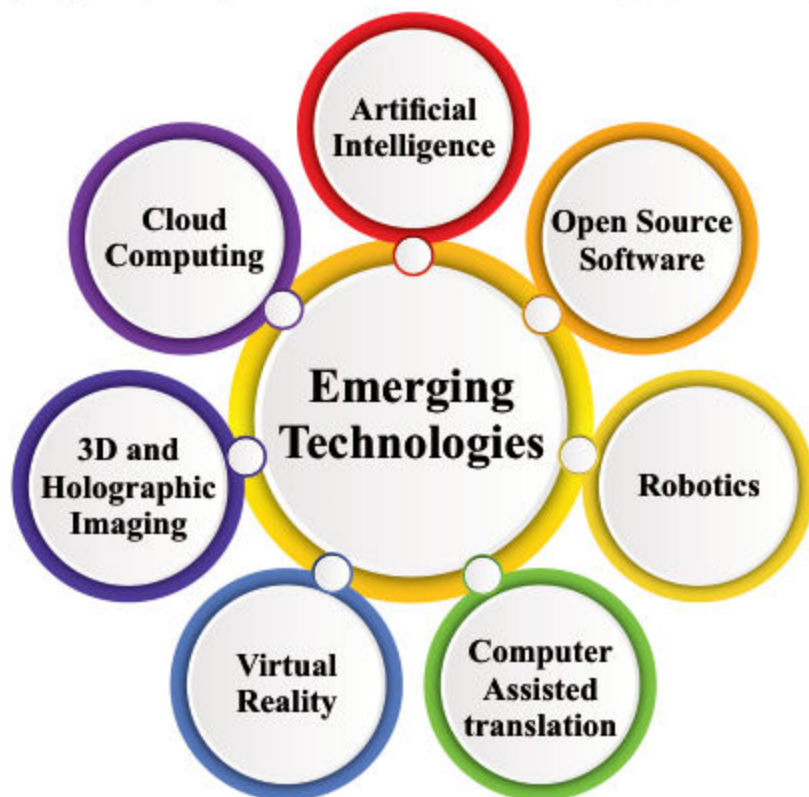


Figure 1.1 Emerging Technologies

The word “Technology” refers to the practical application of scientific knowledge, particularly in the creation of tools and systems that solve problems or improve human life.

1.1.1 Artificial Intelligence (AI):

Artificial Intelligence (AI) helps computers think and learn like people. AI lets machines:

- ◆ Understand what a person says.
- ◆ Learn from mistakes.
- ◆ Choose the best action.

Examples: 1) Self-driving cars that drive without a person (shown figure 1.2),
2) ChatGPT and Gemini that answer questions. Asking Siri or Google for help is AI use in every day life.



Figure 1.2 A Self-driving car

The word Artificial Intelligence was first used by John McCarthy in 1956. He is also known as the “Father of AI”

1.1.2 Robotics:

Robotics is a new technology. It uses programming and sensors. These help machines see and feel what is around them. The machines can choose what to do. They follow instructions like a smart robot toy.

Robots are used in many places, for example:

- 1) In car factories to build cars fast (as shown in Figure 1.3)
- 2) In space to do simple jobs again and again
- 3) In dangerous areas so people stay safe

Example: A robot vacuum cleaner at home sweeps the floor by itself.

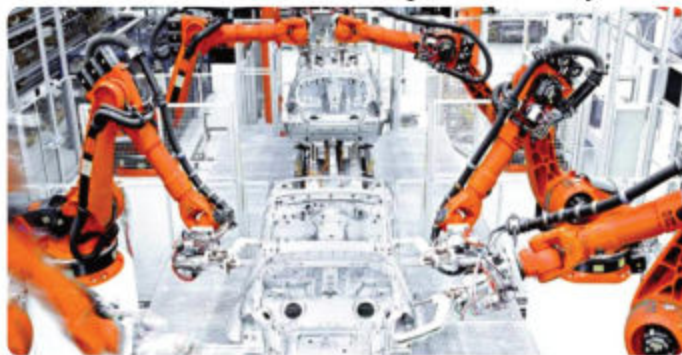


Figure 1.3 Use of robots in auto manufacturing



Activity Time

The Teacher divide the class into two groups A and B and assign group A to make a paper model of a Robot and group B to make a paper model of holographic image viewer .

1.1.3 Biometrics:

Biometrics comes from two words: "Bios" = life and "Metric" = measure. It means measuring parts of human body to know who is the person. It keeps things safe — only authentic person can open.

Examples of body parts used: 1) Fingerprint, 2) Face, 3) Eyes (iris), and 4) Voice.

Biometric is used in daily life for several purposes (see Figure 1.4). For instance: 1) Unlock mobile phone with face or finger, 2) Mark attendance in school with a finger scan.



Figure 1.4
Shows various types of Biometric

1.1.4 Computer-Assisted Translation:

CAT means Computer-Assisted Translation. It is a helper tool for people who change words from one language to another.

What CAT does:

- Helps write the new words.
- Saves old translations to use again.
- Makes work faster.

A real person still thinks and chooses the best words. The computer just helps speed things up.

Famous CAT tools:

Omega T, BasicCAT and Virtaal

Example: A teacher uses CAT to quickly translate an English story into Sindhi.



Figure 1.5 shows the use of computer-assisted translation

1.1.5 Virtual Reality:

Virtual Reality (VR) makes a virtual (artificial) world that feels real. Person wears special glasses or a headset (see Figure 1.6). It feels like person is inside the game or place.

Example: Ali can walk in old Mohenjo-daro! Sara can see the streets, houses, and shops from long ago. It makes person to feel as he/she is really there.

Where VR is used:

- ◆ Learning in school
- ◆ Training (like flying a plane)
- ◆ Games and fun
- ◆ Business meetings in a virtual room

**Figure 1.6**

Shows a student in a classroom learning history through Virtual Reality

1.1.6 3D and Holographic Imaging:

3D Design means making things on a computer. These things have height, width, and depth just like real objects.

Holograms are 3D pictures in the air. A person can walk around and see them from all sides. They use light or laser to shine and look real.

Example:

Draw a dress on paper or computer. Use a hologram app. Watch the dress float in the air like magic (see Figure 1.7).



Figure 1.7 shows a hologram of clothes designed on a computer

1.1.7 Cloud Computing:

Cloud Computing means using the internet to get:

- ◆ Storage space
- ◆ Power to run apps
- ◆ Ready-made software

A person does not need big machines at home. He/She just needs to connect to the internet and use it. A person may have to pay only for what he/she uses like website hosting. There are some free cloud services,

Examples:

- ◆ Save photos in Google Photos
- ◆ Save homework in Google Drive
- ◆ Save games in OneDrive

Open them from any phone, tablet, or computer! No need for USB or hard disk. It's easy, safe, and cheap!



Figure 1.8 shows a child using cloud computing Services

1.1.8 Open Source Software:

Open Source Software means anyone can see and change its code. A person does not need permission from the owner.

Examples:

- ◆ Firefox – a free web browser
- ◆ Java – helps games and apps run
- ◆ Linux – a free computer system

It is like a recipe book everyone can read and improve.



Figure 1.9 Open source software



Activity Time

The teacher divides the class into groups, and each group will look into their homes and school to find examples of common emerging technologies (smartphones, computers, CCTV cameras) and each group makes a list of these examples.

1.2 Progression In Technology:

Technological progress means making better and faster ways to build things. It helps work, talk, and learn in new ways.

Over time, people have improved tools. This has changed everything in daily life.

Two big areas, which are commonly used everyday:

1. Communication – like phones and video calls
2. Computing – like computers and apps.

Example: Old phones were used to make only calls. Now, smartphones do photos, games, and homework.

1.2.1 Progression in Communication Technology (Landline to Smartphones)

This section presents technological progress in communication technology (Landline to Smartphones).

i) Landline phones (1800 to 1990):

Alexander Graham Bell invented the telephone in 1876.

At first:

- ◆ It was only for voice calls.
- ◆ It used wires to connect.
- ◆ Person could not move it.
- ◆ person had to stay close to talk.

Example: A phone at home on the wall or Table Person has to sit near to phone, not walking around. Now phones are wireless and person can walk around while talking.

ii) Mobile Phones (1990 To 2007):

After landline phones, people made mobile phones.

Mobile phones:

- ◆ No wires needed.
- ◆ Person can carry these phones anywhere.
- ◆ Some phones had buttons to press.
- ◆ These phones allowed to play simple games.
- ◆ Person could send text messages.

Example: Old mobile phones were big. Person could play Snake game on them.

iii) Smart Phones (2007 Till present):

Smartphones are used to talk and share today. These smartphones are like tiny computers in human hand.

A person can:

- ◆ Browse the web
- ◆ Take photos and videos
- ◆ Use GPS to find places

- ◆ Watch videos
- ◆ Make video calls
- ◆ Use apps for shopping, learning, and fun

Example: Open YouTube, take a selfie, or play a learning game — all on smartphone.



Figure 1.10 shows the progression in communication technology

1.2.2 Progression In Computer Technology:

Computer progress is split into 5 generations. Each one got better and smaller.

- ◆ First Generation (1940–1956):
 - ◆ Used big vacuum tubes (Figure 1.10).
 - ◆ Computers were huge and heavy.
 - ◆ They used lots of electricity.

Example:

A computer filled a whole room — like a giant box.



Figure 1.11 Vacuum tubes

- ◆ Second Generation (1956–1963):
 - ◆ Used transistors instead of vacuum tubes.
 - ◆ Computers became smaller.
 - ◆ They worked faster.
 - ◆ They used less electricity.

Example:

A computer now fit in a big desk — not a whole room.



Figure 1.12 Transistors

- ◆ Third Generation (1964–1971):
 - ◆ Used Integrated Circuits (ICs).
 - ◆ These were tiny chips with many parts inside.
 - ◆ Computers got much faster.
 - ◆ They became smaller too.

Example:

A computer now fit on a table — like a big box.

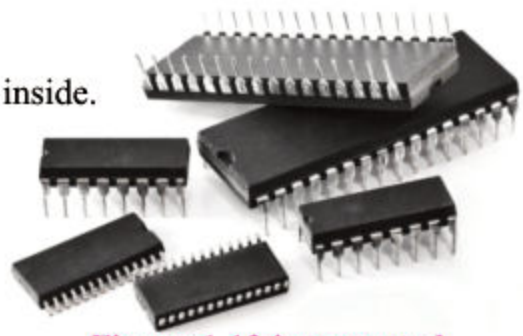


Figure 1.13 intergated circuit(IC)

- ◆ Fourth Generation (1971–now):
 - ◆ Used microprocessors.
 - ◆ These are tiny chips with thousands of circuits.
 - ◆ Computers became super fast.
 - ◆ They got very cheap.

Example:

Nowadays a personal computer is mostly available at home or school. Even a tablet or phone uses this idea.



Figure 1.14 Microprocessor

Fifth Generation (Still in progress) wants real AI in computers. Machines will think and learn like people. Example: Future computers may talk, understand, and help just like a friend.

1.2.3 First Generation Computer Processor to 14th Generation Processor:

CPU means Central Processing Unit. It is the brain of the computer. It does all the thinking and work. New CPU generations are better than old ones. They are faster and smarter.

Companies like Intel and AMD make CPUs. Intel is very famous. Most laptops and desktops use Intel CPUs. Intel has made 14 generations of CPUs. Each one is stronger and quicker.

1st to 14th generations of Intel processors are discussed in Table 1.

Table 1: Timeline of processor generation

Generation	Name(year)	Major change
1 st Generation	Nehalem, 2008-2010	Introduced the Core i series, with improvements in energy efficiency and multi-core processing.
2 nd Generation	Sandy Bridge, 2011	Start of the Turbo Boost 2.0 technology

3rd generation	Ivy Bridge, 2012	Introduced the PCI Express 3.0 standard
4th Generation	Haswell, 2013	New instructions for enhanced performance and improved integrated graphics.
5th Generation	Broadwell, 2014-2015	Smaller chips, lower power consumption and also include security features.
6th Generation	Skylake, 2015	A new microarchitecture with improved performance, power efficiency and new instruction sets.
7th Generation	Kaby Lake, 2016-2017	Advanced video and display capabilities, along with increased clock speeds and additional optimizations for power efficiency
8th & 9th Generation	Coffee Lake, 2017-2019	The addition of more cores improved multitasking
10th Generation	Comet Lake, 2019-2020	Again, adding more cores with faster memory speeds
11th Generation	Rocket Lake, 2021	Improved integrated graphics, start of a new core architecture and support for PCIe 4.0.
12th Generation	Alder Lake, 2021	Hybrid architecture combining performance (P-cores) and efficiency (E-cores) for improved overall performance and power efficiency.
13th Generation	Raptor Lake, 2022-2023	The code name of the 13th-generation Core i-series processors is Raptor Lake. These processors deliver improved multitasking and content creation, higher clock speeds and enhanced gaming performance
14th Generation	Rapoter Lake Refresh 2023-2024	These processors use Intel's new Foveros advanced packaging technology

Usually, when someone buys a computer, the shop person asks: "Which generation do you want?"

It means: "Which Intel CPU generation?"

Example: Intel® Core™ i9-12900H

Let's break it down:

- ◆ Intel → The company
- ◆ Core → The family name
- ◆ i9 → Super powerful
- ◆ 12 → 12th generation
- ◆ H → High speed

Example: A 12th gen i9-H laptop is very fast.



Note:

This method applies only to Intel® Core™ Processors and does not apply to Intel® Pentium®, Intel® Celeron®, or Intel® Xeon®.

1.3 Peripheral devices and their functions:

Peripheral devices are extra helpers for the computer. They connect to the CPU — the brain.

They make computer do more things. Without them, the computer is very limited. A person can connect them with wires or wirelessly (like Bluetooth).

There are 4 main types:

1. **Input Devices:** Put things IN
 - ◆ Keyboard, mouse, webcam
2. **Output Devices:** Show things OUT
 - ◆ Monitor, speaker, printer
3. **Storage Devices:** Save things
 - ◆ USB drive, external hard disk
4. **Networking Devices:** Help connect
 - ◆ Wi-Fi router, modem

Example: Plug in a mouse → ONE can click. Connect a printer → Print a paper.

1.3.1 Input Devices:

Input devices help to put information INTO the computer. They send data or instructions.

Common Input Devices:

- ◆ **Keyboard:** Type letters and numbers
- ◆ **Mouse:** Click and point
- ◆ **Webcam:** Take photos or video
- ◆ **Microphone:** Record voice
- ◆ **Scanner:** Copy paper into the computer

Example: Use the keyboard to type a poem. Use the mouse to open a game. Use the webcam for a video call with friend.



Figure 1.15 Input devices

i) Keyboard

Keyboard is an input device. A person uses it to type:

- ◆ Letters
- ◆ Numbers
- ◆ Symbols

It has special keys too:

- ◆ **Arrow keys:** Move up, down, left, right
- ◆ **Navigation keys:** Go through menus

What a person/user can do:

- ◆ Scroll in a story
- ◆ Move the cursor (blinking line)
- ◆ Open menus in games or apps

Example: Press arrow keys to move in a game.

ii) Mouse

Mouse is an input device. It helps to control the computer. It has buttons and scroller (see Figure 1.15).

5 things a mouse can do:

1. Select - Click on a picture
2. Navigate - Open menus
3. Move cursor - Point to words
4. Drag and drop - Move icons
5. Scroll - Go up and down pages

How it connects:

- ◆ With a wire (plugged in)
- ◆ Or wirelessly (Bluetooth)

Example: Use the mouse to drag a photo into a folder. Scroll to read a long story.

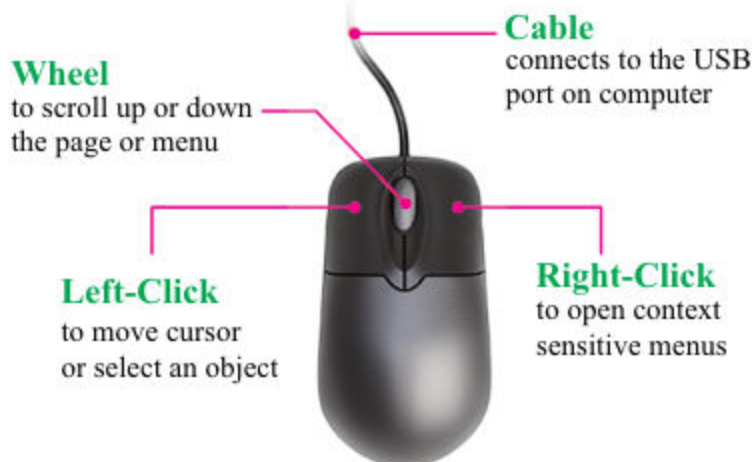


Figure 1.16 shows a computer mouse

iii) Webcam

Webcam is an input device. It takes pictures or videos. It turns them into digital files with software. Nowadays, it is integrated in laptops (see Figure 1.16).

How it connects:

1. With a USB cable
2. Or built inside the computer/laptop

What a person uses it for:

- ◆ Chatting with friends
- ◆ Live streaming games
- ◆ Recording videos
- ◆ Video calls in class

Example: Turn on the webcam and wave to friend on a video call.



(a)



(b)

Figure 1.17 shows an external webcam (a) shows screen-integrated webcam

iv) Scanner:

Scanner is an input device. It uses light to copy things.

It turns paper into digital files. This keeps them safe and easy to find.

Common Scanners:

1. Barcode Scanner

- ◆ Used in shops
- ◆ Scans price tags fast



2. Optical Scanner

- ◆ Used in fax machines
- ◆ Copies pictures and drawings



3. Flatbed Scanner

- ◆ Has a lid that opens
- ◆ Put paper inside
- ◆ Used in offices for books or photos



Example:

Scan drawing paper at school; Save it on the computer. Share it with teacher.

1.3.2 Output Devices:

Output device is a helper. It takes information from the computer. It turns it into something a user can see or hear.

Common Output Devices:

- ◆ **Monitor** - Shows pictures and words
- ◆ **Printer** - Makes paper copies
- ◆ **Speakers** - Plays sounds and music
- ◆ **Projector** - Shows big images on a wall

Example:

Watch a video on the monitor. Print homework with the printer. Listen to a song with speakers.

i) Monitor:

Monitor is the main output device. It shows pictures and words on a screen.

It gets signals from the computer. A graphics card helps it understand. Then it displays the image or video.

Example:

A laptop screen is a monitor. Watch cartoons on it.

CRT Monitor:

- ◆ Stands for Cathode Ray Tube
- ◆ It is the oldest type of computer screen
- ◆ Uses a big vacuum tube inside
- ◆ Makes pictures with electron beams



LCD Monitor:

- ◆ Stands for Liquid Crystal Display
- ◆ It is a flat screen
- ◆ Uses liquid crystals and electric signals
- ◆ Makes clear pictures



LED Monitor:

- ◆ Stands for Light Emitting Diode
- ◆ Uses tiny lights called LEDs
- ◆ It is very thin
- ◆ Uses less power
- ◆ Shows bright and real colors



OLED Monitor:

- ◆ Stands for Organic Light Emitting Diode
- ◆ Uses special glowing materials
- ◆ It is more expensive than LED
- ◆ Uses very little power
- ◆ Shows super clear and bright pictures



ii) Printer:

Printer is a popular helper device. It prints text and pictures on paper. It makes hard copies from the computer.

How it connects:

- ◆ **Wi-Fi** - No wires (wireless)
- ◆ **USB (Universal Serial Bus) cable** - Plugged in

What it can print:

- ◆ Homework
- ◆ Reports
- ◆ Letters
- ◆ Photos

Example: Click Print and get drawing/letter on a paper.

There are many types of printers as shown in Figure 1.14

➤ **Laser Printer**

They use laser beam for printing. These printers are fast and cost-efficient

➤ **Inkjet Printer**

produces prints by spraying tiny droplets of liquid ink onto paper. These printers produce high-quality images.

➤ **Receipt Printer**

They use heat for printing. They are fast, quiet, and low-maintenance.

Figure 1.14 shows different types of printers

iii) Speaker:

Speakers are output devices. They turn computer signals into sound.

Types:

- ◆ **Built-in speakers**
Inside laptops and desktops (see Figure 1.18)
- ◆ **External speakers**
Connect with USB or audio jack

What a person can use them for:

- ◆ Listen to music
- ◆ Watch movies
- ◆ Join online classes

Example:

Play a song or hear your teacher in a video call!



Figure 1.18
Laptop internal speakers

iv) Projector:

Projector is an output device. It shows big pictures from the computer. It puts the image on a wall or screen.

How it connects:

- ◆ **HDMI (High-Definition Multimedia Interface) cable**
it is used for clear video (see Figure 1.19)
- ◆ **VGA (Video Graphics Array) cable** - Older type

Where a person can use it:

- ◆ **School** - Show lessons
- ◆ **Office** - Meetings
- ◆ **Home** - Watch movies

Example: A teacher uses a projector to show a story on the big wall.



Figure 1.19 shows the Projector and the HDMI cable

1.3.3 Storage devices:

Storage devices save and get back data. They help computers keep lots of files.

Common Storage Devices:

- ◆ **USB drive**
Small stick you plug in
- ◆ **External hard drive**
Big box with cable
- ◆ **SSD (solid-state drive) drive**
Fast and inside modern computers

Example:

Save homework on a USB drive. Take it to school and open it there.

i) USB Flash/Pen Drive:

USB Drive (also called pen drive) is a small storage tool. It plugs into the USB port of a computer (see Figure 1.20).

What it does:

- ◆ Saves files
- ◆ Backs up data
- ◆ Moves files from one device to another

Example:

Save a project on a USB drive. Take it to school and plug it in computer/laptop.



Figure 1.20
Different types of USB drives

ii) External Hard Drive:

External Hard Drive is a portable storage box (See Figure 1.21). It connects to a laptop or PC (Personal Computer) with a USB cable.

Main Advantage:

- ◆ Gives extra space for files
- ◆ More room than the built-in drive

Example:

Save all photos, videos, and games. Carry it anywhere and plug it in computer/laptop.



Figure 1.21 shows an External drive connected to the Laptop through the USB port

iii) SSD Drive:

SSD means Solid State Drive. It is a fast storage device (see Figure 1.22).

Why it's better:

- ◆ More space
- ◆ Super speed
- ◆ Uses less power

It works using special silicon chips. SSDs are newer and smarter than old drives.

Example:

Games load in seconds on a computer with SSD.



Figure 1.22 SSD drive

1.3.4 Network Devices:

Network devices help computers talk to others. They share data over a network or the Internet.

Common Network Devices:

- ◆ **Modem** - Connects to the internet
- ◆ **Wireless Access Point** - Makes Wi-Fi
- ◆ **Network Adapter Card** - Lets devices join the network

Example: Use a modem to go online. Connect with Wi-Fi to share files.

i) Modem

Modem helps to get internet. It changes signals such that computers can be connected to the Internet (see Figure 1.24).

How it works:

- ◆ Turns computer data into phone line signals
 - ◆ Turns phone signals back into computer data
- It is a translator between computer/device and the ISP.

Example: Plug in the modem. Open YouTube. Watch videos fast.



Figure 1.24 A modem with an Ethernet cable connector

ii) Wireless access point:

WAP means Wireless Access Point. It helps wireless devices connect to a wired network (see Figure 1.25).

Why use it:

- ◆ No cables needed
- ◆ Easy to set up
- ◆ Connects all devices in one place

Example: At home Wi-Fi router is a WAP. Phones, laptops, and tablets join the internet without wires. **shows a wireless access point device**



Figure 1.25

iii) Network interface card(NIC):

Network Adapter Card helps computer send and receive data. It connects to other devices on the network.

It is fixed on the motherboard inside the computer.

Example: Use it to connect to Wi-Fi or play online games.

Without it, computer cannot be connected to the Internet.



Figure 1.26 Shows a network interface card



Activity Time

The teacher will make a Venn diagram on the board with two overlapping circles labeled as “modem” and “NIC Card” respectively then ask student to fill these circle with function of both devices so they can differentiate between their functions and find similarities .

1.4 I/O Ports and their functions:

Ports are small holes on a computer. They connect helper devices (peripherals).

What do ports do:

- ◆ Send data from CPU to devices
- ◆ Get data from devices to CPU

Where these ports are present:

- ◆ Back of desktop computer (see figure 1.27 a)
- ◆ Side of laptop (see figure 1.27 b)

Common Ports:

- ◆ **USB** - For mouse, keyboard, pen drive
- ◆ **HDMI** - For TV or monitor
- ◆ **Audio jack** - For headphones
- ◆ **Power port** - To charge

Example: Plug mouse into a USB port. It works immediately.



Figure 1.27

- (a) Shows external ports on the back of the Desktop.
 (b) Shows external ports on the side of the laptop

i) SATA Ports:

Serial Advanced Technology Attachment (SATA) Port connects hard disks and SSDs (Solid State Drives) to the motherboard. It is inside the computer (see figure 1.28).

What does SATA Port do:

- ◆ Sends data from CPU to disk
- ◆ Gets data from disk to CPU

Example: A game and photos load fast because of SATA.

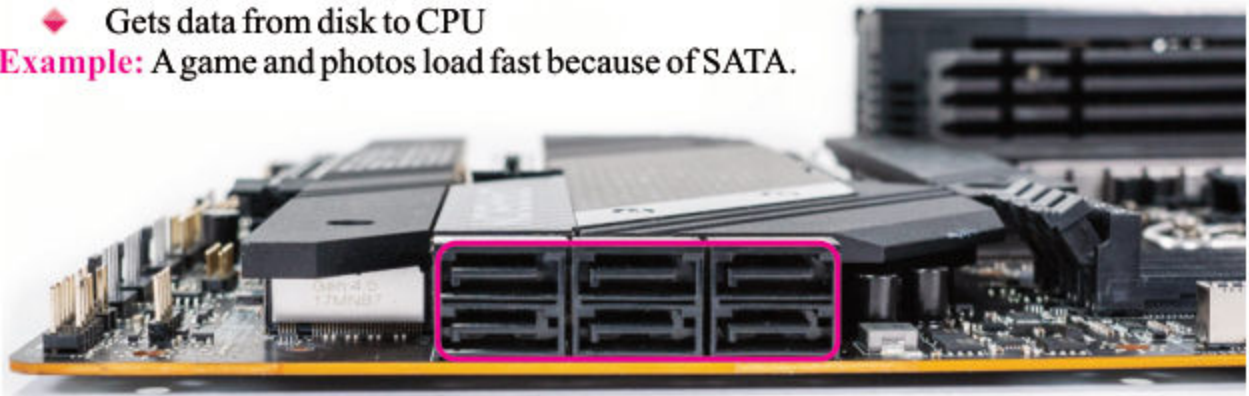


Figure 1.28 shows SATA ports

ii) PCI & PCI Express (PCIe) Port:

PCI (Peripheral Component Interconnect) Port is used in desktop PCs. It is a fast slot on the motherboard (see figure 1.29).

PCI Port connects extra cards like:

- ◆ Graphics card (for games)
- ◆ Sound card
- ◆ Network card

PCI-e is a newer and faster version. It replaces old PCI. It is also a slot inside the computer.

Example: A gaming PC has a graphics card in a PCI-e slot. It makes games look smooth and colorful.

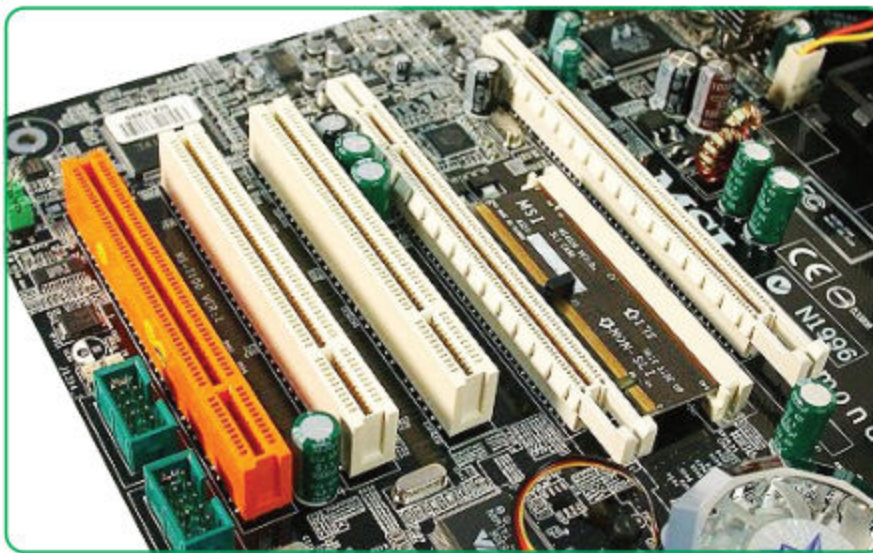


Figure 1.29 shows the PCI expansion port

iii) Serial Port:

A serial port sends data one bit at a time in order. It has 9-pin or 25-pin connectors (see figure 1.30). It is also called a COM port (for example, COM1 or COM2).

Its uses:

- ◆ Connecting modems
- ◆ Connecting microphones
- ◆ Connecting older printers

Example: In old computers, a mouse or modem was connected using a serial port.

Note: Today, USB ports are used instead of serial ports.



Figure 1.30 Serial port

iv) Parallel Ports:

A parallel port sends 8 bits (one byte) at a time. It is faster than a serial port. It is also called a printer port (see figure 1.31).

Its uses:

- ◆ Connecting printers
- ◆ Connecting scanners

Example: Old computers used parallel ports to attach printers.

Note: Today, USB ports have replaced parallel ports.



Figure 1.31 shows a parallel port

v) USB Port:

USB stands for Universal Serial Bus. The first non-commercial USB port came out in 1994.

Its uses:

- ◆ Connect mouse, keyboard, printer, scanner, external hard disk
- ◆ Transfer data quickly

Its connector:

- ◆ The oval-shaped plug a person puts into phones or computers.

Types of USB Ports (see figure 1.32):

- ◆ Type-A – Common rectangle shape
- ◆ Type-B – Square with curves
- ◆ Type-C – Small, reversible
- ◆ Micro USB – Tiny, used in old phones

Example: Plug a pen drive into a Type-A port to copy homework.

USB Type-A



USB Type-B



USB Type-C



Micro USB



Figure 1.32 shows different USB ports with their connectors

vi) HDMI port:

HDMI means High-Definition Multimedia Interface (see figure 1.33). It connects computer to TV, monitor, or projector. It also sends clear video and sound using one cable.

Its uses:

- Extend laptop screen
- Mirror display
- Play movies or give presentations

Example: Plug HDMI cable from laptop to TV. Watch a game or video on a big screen.

Note: One thin cable does everything.



Figure 1.28 HDMI Port

vii) Ethernet port:

Ethernet port (also known as an RJ45 port) helps to plug in an Ethernet cable (also called a network cable). It connects computer to a network or Wi-Fi router (see figure 1.34). It is a small rectangular hole with eight metal contacts. It is, usually, labeled with a network icon or “LAN” on backside of desktop computer and side of laptop.

Its benefits:

- More stable than Wi-Fi
- Usually faster internet

Example: Plug cable into Ethernet port; Play online games using the Internet.



Figure 1.2 Ethernet Port and cable



Summary

- ◆ Emerging technologies are innovations that have the potential to create a new industry or transform an existing one.
- ◆ A peripheral is an external hardware device that gives a computer additional functionality.
- ◆ Input devices send information to computer system.
- ◆ Output devices are devices that convert computer data into a human-understandable form, such as text, images, or sound.
- ◆ Storage devices are devices that store data. It allows data to be saved, retrieved, and transferred as needed.
- ◆ Ports are physical connection points on a computer where user can plug in external devices.



Terms to Remember

AI	Using computers to do things that traditionally require human intelligence is called Artificial Intelligence(AI)
VR	Virtual Reality is defined as the use of computer simulation that allows a human to be immersed in an artificial 3D environment.
HDMI	High Definition Multimedia Interface is a port used to connect a monitor and projector to computer.
USB	Universal Serial Bus ports are uses to connect devices, transfer data and charge electronic devices.
CRT	Cathod ray tube is a display that use a vacuum tube and electronic gun to produce images on its screen.
LED	An LED monitor, or Light-Emitting Diode monitor, is a type of display device that uses LED technology to produce images.
OLED	An OLED monitor is a type of display device that uses Organic Light-Emitting Diode technology to produce images

EXERCISE

1. Encircle the correct answer:

- i. Through 3D image processing we can:
 - (a) Show Hologram
 - (b) Move items automatically in a factory.
 - (c) Print a realistic image
 - (d) Recognize patterns and make decisions
- ii. Biometric authentication can be done through:
 - (a) Our account number.
 - (b) Typing password.
 - (c) Using Face, Fingerprint and voice.
 - (d) By phone number.
- iii. Which of the following is not classified as an emerging technology?
 - (a) Self-driving car
 - (b) Robotics.
 - (c) Banking.
 - (d) Artificial Intelligence.
- iv. Which input/output port is most commonly used for connecting devices like USB drives, a mouse, a printer and a keyboard:
 - (a) Serial port
 - (b) USB port.
 - (c) HDMI port.
 - (d) Parallel port
- v. The main purpose of peripheral devices is
 - (a) To give additional functionality to the computer
 - (b) To store files.
 - (c) To send/receive data from external devices.
 - (d) All of the above.

2. Fill in the blanks with appropriate given words:

Ports, Peripheral, HDMI, Input, 1994

- i) Devices like mouse and keyboard are refer as _____ devices.
- ii) _____ port is used to connect projector to computer.
- iii) _____ devices are those devices that are used to give input and take output from the system unit.
- iv) _____ are the physical connections on the front or back of the computer case that allow us to plug in the I/O devices.
- v) The first non-commercial USB port was released in _____.

3. Provide descriptive answers of the following questions.

- i) Describe how emerging technologies are changing this world?
- ii) Write how the application of VR helps in education?
- iii) Briefly describe how progression in technology affects daily life?
- iv) Write the difference between printer and scanner?
- v) Describe why peripheral devices are necessary for a computer?



Class Activity

The teacher takes the students to the lab where she/he should demonstrate to them how to plug in different peripheral devices through I/O ports of computer and fill the worksheet given below.

Peripheral device name	I/O port Name



Instructions for Teachers

Teaching emerging technologies to children is very thought-provoking but if you give them real-life examples, it will become easy to understand

- Encourage children to create a wall in your classroom with the heading of “different input and output devices “. As the chapter progresses, add a new picture on the wall.
- Divide them into teams so that they learn to do web searches on emerging technologies.
- Help them install device driver in lab, hence they can have practical exposure to installing external hardware.

Chapter

2

DIGITAL SKILLS

Student Learning Outcomes:

After the completion of this unit students will be able to:

- ◆ Understand basics of Word processing using MS Word
- ◆ Explore functions of MS Word
- ◆ Explore functions of Presentation Software using MS Power Point
- ◆ Understand the basics of electronic mailing (Email)
- ◆ Demonstrate usage of Email

